

Data Mining

Intro and Course Topic Overview

Howles

What is Data Mining?

- Sometimes called data or knowledge discovery
- Finding *true* patterns while avoiding overfitting (finding false patterns due to randomness)

Defining Data Mining

- Data mining has many different definitions, sometimes dependent on your background (math, statistics, database, visualization, machine learning). In other words, it is inherently multidisciplinary in nature
- Previously known as data fishing, dredging
- DM also known as knowledge discovery (KDD); often used interchangeably

Our Definition

- For our purposes, we will define data mining (in its simplest form) as the process of looking for previously *unknown, valid, interesting* patterns in data
- Hopefully, the patterns discovered are potentially useful and ultimately understandable
- It is important to remember that DM results are seldom static; they may age and become obsolete
- Also important: strong patterns → good prediction

Data Mining –

HIGHLIGHT THIS SLIDE

- Requires *human intervention* to verify model integrity
- Requires *careful interpretation* of the results – Do they make sense?
- Requires *multiple iterations* to find good models
- Is full of surprises – every data set contains its own unique problems

More Observations

- Data mining involves using historical data – your models are only as good/accurate as your historical data
- It is important to remember that DM results are seldom static; they may age and become obsolete
- Having a forensic mind is helpful

Even more observations

- Data mining reveals correlations, not causations
- You may be able to obtain some insight into the cause but you may not really know why
- You need to be careful that you don't try too hard – remember the joke about statisticians!

Data Mining isn't

- What is a database?
- In a database, describe a *query*
- What is the difference between a database query and data mining?

Origins of Data Mining

- Draws from machine learning, AI, pattern recognition, statistics and database systems
- Generally two types: Predictive and descriptive methods
- Predictive – predict an unknown value
- Descriptive – find patterns that describe the data

DM in practice

- A farmer needs to determine which cows to trim from his herd
- A physician needs to identify healthy cells
- A system administrator needs to predict CPU performance under certain circumstances
- A florist needs to classify flowers

Trimming the herd

- This sounds rather trivial
- But consider that cows are described by ~ 700 features
- How do you determine good picks?

Challenges

- Need software to help
 - Lots of data
 - Patterns may not be obvious
 - Data? May not be “clean”, available
 - May not have access to needed data

Who's Using Data Mining?

- Direct marketing – said to be “The” most established users of data mining
 - Response modeling, cross-selling
- Telecommunications
 - Churn reduction, fraud detection, cross-selling

Who's Using Data Mining (2)

- Finance
 - Credit scoring, loan approvals, fraud detection
- Governments
 - Tax compliance, fraud detection
- Insurance
 - Fraudulent claim detection, customer acquisition and marketing

Who's Using Data Mining (3)

- Non-Profits
 - Donations, mail targeting
- Stock Market
 - Stock timing, fund management
- Pharmaceuticals
 - Drug discovery, combinatorial chemistry
- Sports teams
 - Optimizing rosters, scoring tendencies

Who's Using Data Mining (4)

- Engineering
 - Signal detection, fault diagnostics
- Computer Systems and Security
 - Intrusion detection
- Retail
 - Market basket analysis, target mailing – beer/diaper syndrome (Blischok)
- Credit: Dean Abbott

Challenges

- Need software to help:
 - Lots of data
 - Patterns may not be obvious
 - Data? May not be clean or available
 - May not have access to needed data

Problems with Patterns

- Many patterns are not interesting
- Patterns may be inexact
- Data may be garbled or missing

But, Patterns can be ...

- Used to predict outcomes in new or different situations
- Used to understand and explain how prediction is derived

Historical Perspective

- Data mining is a relatively new field but the technology has been around for years
- What may be considered ‘new’ are the algorithms and methods – combined with dramatically increasing hardware storage and speed abilities

The Need for Data Mining

- “We are data rich but knowledge poor” (Source: Unknown)
 - What does this mean?
 - Data quantity doubles every 20 months (Witten & Frank, 2000)
- Data used to be stored in filing cabinets or a desk drawer – now electronic
- Data/Information in millions of categories being stored by millions of people, companies and entities

Learning

- What is learning? What does it mean to learn something?
- How do we assess learning?
- How do we do this with a machine?
- What does machine learning really mean?

Learning

- Often described as the process of acquiring knowledge.
- How do we acquire knowledge? Study, are taught, observe, memorize, receive instruction
 - This sounds very human

Assessment

- How do we assess learning?
- Testing, demonstration

Machine Learning

- ML introduces many legal and philosophical issues
- We won't get into them much in this course
- We will just accept the fact that machines can “learn” (using the term very loosely)
- We will focus on the practical aspect of what is happening and how

Definitions

- Data
- Information
- Knowledge
 - 36 23 12 -1 36 24 41 – this is data
 - The numbers represent temperature readings in Rochester during November, 2008 (this is information; it has a context)
 - Knowledge involves interpreting the information to determine that I need to wear a coat in Rochester during November

More Definitions

- Attributes – features that may or may not be interesting (to us)
 - Examples: temperature, age, cost
- Instance – a logical grouping of attributes (we often refer to as a record)
- Dataset – a collection of instances (the file itself)
- Concept – classification, association, clustering or numeric prediction; what can be learned

How do we categorize data mining problems?

- What we are trying to learn
- The task

Supervised and Unsupervised

- We typically categorize DM algorithms as supervised or unsupervised
- Supervised: Algorithm is provided info about what group or class to which each instance belongs
- Unsupervised: Algorithm is not provided group/class info

Supervised Examples

- Online-line analytical processing
- Neural networks
- Genetic algorithms
- Rule induction
- Decision trees
- K-nearest neighbor
- Classification

Unsupervised Examples

- K-means (clustering)
- Bayesian Belief Nets
- Association rules

Data Mining Tasks

- Classification – predicting where an item belongs (wrt a class)
- Clustering – finding clusters in the data
- Associations – A & B occurs with C frequently
- Visualization – making it easier for humans to understand or interpret
- Summarization – descriptive analysis
- Deviation Detection – finding changes
- Estimation – predicting a continuous value
- ...

Forms of Data Mining

- We' ll look at these:
 - Classification (predictive)
 - Association (descriptive)
 - Clustering (descriptive)
 - Numeric prediction (predictive)
 - Text Mining

Classification

- You have a dataset with known criteria and specific outcome classifications. You receive a new instance that has not been seen before. To which class does it belong?
- The classification is based on patterns and relationships seen before.
- A form of supervised learning because the classes are already known – you are just trying to see where a new instance fits

Classification Methods

- Linear regression – do the data points match a line? Tries to fit the data to what is known
- Decision trees
- Neural Networks – can be more accurate but easy to overfit the data (find patterns in random noise)

Classification Examples

- Who is likely to purchase a certain product?
- Who is at high risk of acquiring an illness?
- Which student projects are worth a passing grade?
- Which objects on a radar screen are vehicles?
People?

Association

- Predicting a class or missing attributes
- Determined by rules, rules may be weighted (all rules are not the same value or importance)
- Look for associations between features
- Market Basket Analysis

Clustering

- Finding groups of items that are similar
- Clustering is *unsupervised* because the classifications are not known
- Data items are grouped according to logical relationships. Example: identify market segments, identify flowers

Numeric Prediction

- Unlike previous examples, numeric prediction is .. Well, predicting a numeric value
- Outcome is not a classification but a specific value
- Based on history, trends, other data values

Text Mining

- Finding patterns in text
- Any text: Documents, web pages
- Data may be
 - Structured
 - Data provided on a survey
 - Unstructured
 - Text in a book or paper
 - Semistructured
 - Email messages

Problem - Overfitting

- Overfitting means
 - May occur if we follow the training data too closely
 - Build a decision tree with too much detail; doesn't necessarily become a better classifier
- Overtraining
- Forcing results

Data Cleaning

- An important aspect of DM
- Handling invalid values, duplicates, missing data, data entry errors
- If necessary, converting data to specific values in order to perform correct measurements (putting a value to good, bad, healthy ..)

Data Cleaning

- One of the most important and time consuming tasks
- If not well, it can compromise the entire project
- We will talk in detail about cleaning

Ethics and society

- Many important social and ethical issues surround DM
- Privacy – DM can reveal a lot of hidden information about individuals. Is it our business? Is it a violation of others' privacy?
- Ethics is a critical concern not always given the time and attention it requires

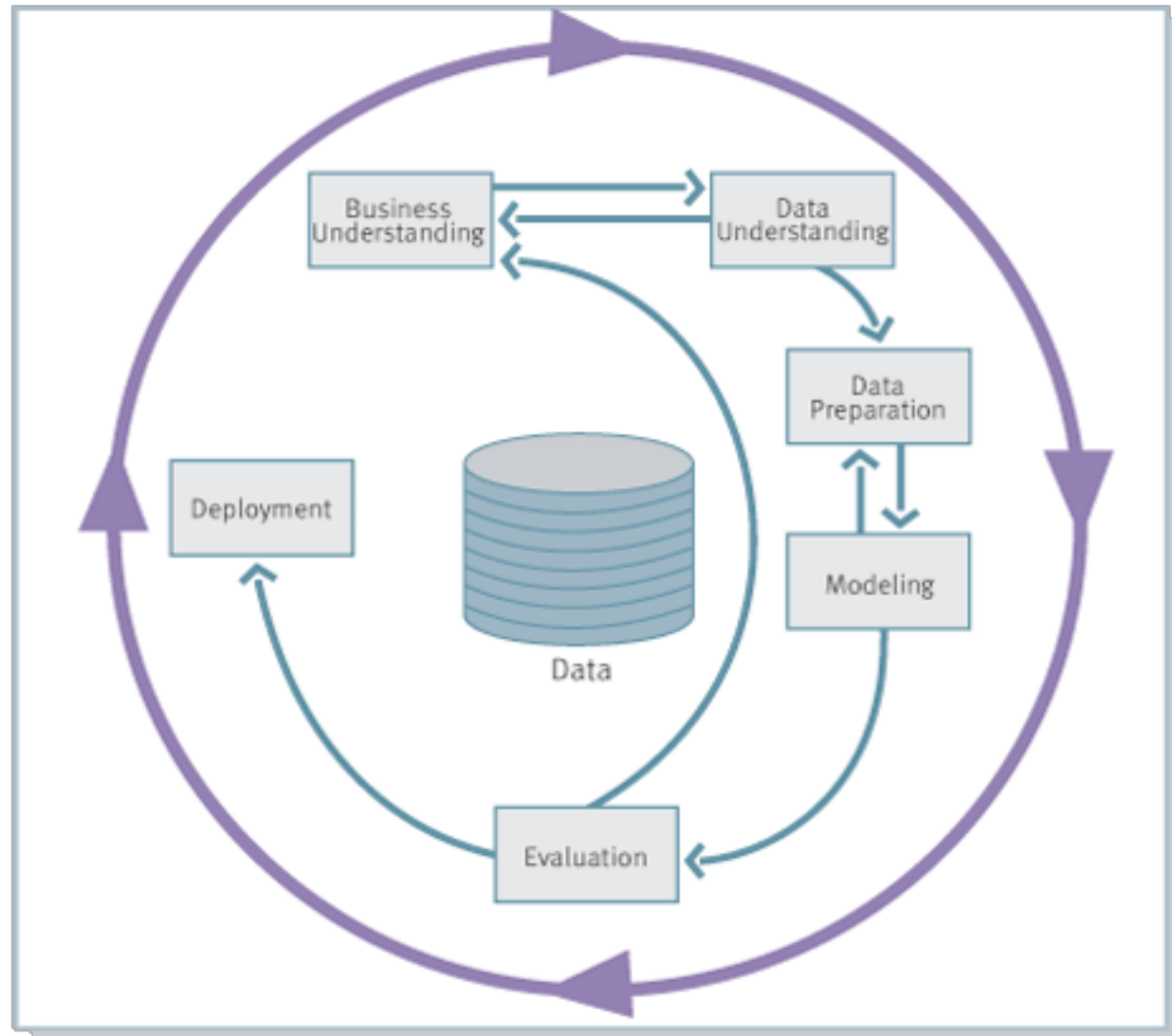
Other Issues

- Data integrity – analysis is only as good as the data. How can we ensure the data is valid and complete?
- Cost – Continuing pressure for larger and faster systems. Driven by the quantity of data and the need for results in a limited amount of time

CRISP

- Understanding the business is similar to the requirements step in the software development cycle – if you don't understand the problem, know the available resources and know what the customer is expecting, you are destined to fail

CRISP



The Process

- CRISP-DM
- Data is at the center (of course!) but DM doesn't make sense unless you understand the business (context) and the data
- To be successful, you need data, domain and data mining experts
- Data preparation is an important step
- Evaluation (do the results make sense)?

References and Credits

- Dr. Carol Romanowski, RIT
- Dr. Andrew Moore, CMU
- Witten & Frank, *Data Mining Practical Machine Learning Tools and Techniques* (2005)
- Max Bramer, *Principles of Data Mining* (2007) Springer.